

# Complete FHIR Inpatient (IP) Workflow Guide

## Overview

This document explains how a complete inpatient (IP) workflow works using FHIR resources.

This guide is designed for:

- Hospital Information Systems (HIS)
- EMR/EHR systems
- Bed management systems
- Admission/Discharge/Transfer systems (ADT)
- Nursing workflows
- Clinical workflows
- Billing workflows
- Insurance workflows
- AI workflow orchestration
- Queue management systems

This document provides:

- Real-world inpatient workflow
- FHIR resource relationships
- Step-by-step architecture
- Detailed JSON examples
- Bed/ward management concepts
- Admission workflows
- Transfer workflows
- Discharge workflows
- Billing and insurance flow
- AI orchestration ideas

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## What is Inpatient Workflow?

Inpatient means:

Patient is admitted into the hospital  
and stays for ongoing care.

Examples:

- Surgery admission
- ICU admission
- Observation admission
- Cardiology admission
- Maternity admission
- Emergency admission

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## OUTPATIENT vs INPATIENT

| Outpatient               | Inpatient                 |
|--------------------------|---------------------------|
| Short visit              | Long stay                 |
| No admission             | Admitted                  |
| Usually one encounter    | Multiple encounter stages |
| Simple workflow          | Complex workflow          |
| Minimal bed tracking     | Bed/ward management       |
| Lower billing complexity | Higher billing complexity |

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## STRONGLY RECOMMENDED RESOURCES

For inpatient systems, these become VERY important.

| Resource           | Priority |
|--------------------|----------|
| Encounter          | CRITICAL |
| Location           | CRITICAL |
| Task               | CRITICAL |
| Coverage           | HIGH     |
| AllergyIntolerance | HIGH     |
| Medication         | HIGH     |
| CarePlan           | HIGH     |
| DocumentReference  | HIGH     |

| Resource       | Priority  |
|----------------|-----------|
| Provenance     | HIGH      |
| AuditEvent     | HIGH      |
| EpisodeOfCare  | VERY HIGH |
| ServiceRequest | VERY HIGH |
| Procedure      | VERY HIGH |

## IMPORTANT ARCHITECTURAL CONCEPT

### EpisodeOfCare

Strong recommendation.

An inpatient admission is usually:

One EpisodeOfCare  
containing  
multiple Encounters

### Example

```
Hospital Admission
  ↓
EpisodeOfCare
  ├── ER Encounter
  ├── Admission Encounter
  ├── ICU Encounter
  ├── Ward Encounter
  └── Discharge Encounter
```

## HIGH LEVEL INPATIENT FLOW

```
graph TD; A[Patient Registration] --> B[Admission Decision]; B --> C[Bed Allocation]; C --> D[Admission Encounter]; D --> E[Nursing Assessment]; E --> F[Doctor Evaluation]; F --> G[Orders (Labs/Imaging/Procedures)]; G --> H[Medication Administration]; H --> I[Monitoring & Progress Notes]; I --> J[Transfers (Ward/ICU)]; J --> K[Discharge Planning]; K --> L[Final Billing]; L --> M[Insurance Claim]; M --> N[Discharge];
```

## CORE RESOURCE RELATIONSHIP MAP

```
graph TD; A[Patient] --> B[EpisodeOfCare]; B --> C[Encounter]; C --- D[Location]; C --- E[Observation]; C --- F[Condition]; C --- G[ServiceRequest]; C --- H[Procedure];
```

- └─ MedicationRequest
- └─ DiagnosticReport
- └─ DocumentReference
- └─ Invoice
- └─ Claim
- └─ Task

## STEP 1 — Patient Registration

### Resource

- Patient

### Example

```
{
  "resourceType": "Patient",
  "id": "patient-1",
  "active": true,
  "name": [
    {
      "text": "Rahul Sharma"
    }
  ],
  "gender": "male",
  "birthDate": "1988-05-10"
}
```

## STEP 2 — Insurance Verification

Strong recommendation:

Use:

Coverage

## Coverage Example

```
{
  "resourceType": "Coverage",
  "id": "coverage-1",
  "status": "active",
  "beneficiary": {
    "reference": "Patient/patient-1"
  },
  "payor": [
    {
      "display": "ABC Insurance"
    }
  ]
}
```

---

## STEP 3 — Admission Decision

Doctor decides patient requires admission.

Common reasons:

- Surgery
- Observation
- Critical monitoring
- ICU care
- Severe infection

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## STEP 4 — Create EpisodeOfCare

### WHY IMPORTANT

Tracks the complete hospitalization.

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### Example

```
{
  "resourceType": "EpisodeOfCare",
```

```
{
  "id": "episode-1",
  "status": "active",
  "patient": {
    "reference": "Patient/patient-1"
  },
  "type": [
    {
      "text": "Inpatient Admission"
    }
  ],
  "managingOrganization": {
    "reference": "Organization/org-1"
  }
}
```

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## STEP 5 — Bed & Ward Allocation

VERY IMPORTANT.

Use:

Location

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## Recommended Location Hierarchy

```
Hospital
├── Building
├── Floor
├── Ward
├── Room
└── Bed
```

---

## Example

```
{
  "resourceType": "Location",
```

```
"id": "bed-101",
"status": "active",
"name": "Ward A - Bed 101",
"physicalType": {
  "text": "Bed"
}
}
```

---

## STEP 6 — Create Admission Encounter

### Resource

- Encounter

## IMPORTANT

Use:

```
"class": {
  "code": "IMP"
}
```

Meaning:

Inpatient Encounter

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## Example

```
{
  "resourceType": "Encounter",
  "id": "encounter-admission-1",
  "status": "in-progress",
  "class": {
    "code": "IMP"
  },
  "subject": {
```

```

    "reference": "Patient/patient-1"
  },
  "episodeOfCare": [
    {
      "reference": "EpisodeOfCare/episode-1"
    }
  ],
  "location": [
    {
      "location": {
        "reference": "Location/bed-101"
      }
    }
  ]
}

```

## STEP 7 — Admission Assessment

Nurse and doctor perform assessment.

Resources:

- Observation
- Condition
- QuestionnaireResponse
- DocumentReference

## Nursing Assessment Example

```

{
  "resourceType": "Observation",
  "id": "obs-admission-1",
  "status": "final",
  "code": {
    "text": "Admission Assessment"
  },
  "subject": {
    "reference": "Patient/patient-1"
  },
  "encounter": {
    "reference": "Encounter/encounter-admission-1"
  }
}

```

```
    },  
    "valueString": "Patient conscious and stable"  
  }  
}
```

---

## STEP 8 — Diagnosis Recording

### Resource

- Condition

---

### Example

```
{  
  "resourceType": "Condition",  
  "id": "condition-1",  
  "clinicalStatus": {  
    "text": "active"  
  },  
  "code": {  
    "text": "Pneumonia"  
  },  
  "subject": {  
    "reference": "Patient/patient-1"  
  },  
  "encounter": {  
    "reference": "Encounter/encounter-admission-1"  
  }  
}
```

---

## STEP 9 — Create CarePlan

Strong recommendation.

CarePlan coordinates:

- medications
- procedures
- monitoring
- goals

- discharge planning

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## Example

```
{
  "resourceType": "CarePlan",
  "id": "careplan-1",
  "status": "active",
  "intent": "plan",
  "subject": {
    "reference": "Patient/patient-1"
  }
}
```

---

## STEP 10 — Doctor Orders

Use:

ServiceRequest

Examples:

- CBC
- MRI
- Surgery
- ICU transfer
- Physiotherapy

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## Example

```
{
  "resourceType": "ServiceRequest",
  "id": "sr-cbc-1",
  "status": "active",
  "intent": "order",
  "code": {
    "text": "Complete Blood Count"
  }
}
```

```
    },  
    "subject": {  
      "reference": "Patient/patient-1"  
    }  
  }  
}
```

---

## STEP 11 — Task Orchestration

VERY IMPORTANT.

Inpatient systems REQUIRE Task.

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### Example Workflow

```
Admission Task  
  ↓  
Vitals Task  
  ↓  
Lab Collection Task  
  ↓  
Medication Administration Task  
  ↓  
Doctor Review Task  
  ↓  
Discharge Planning Task
```

---

### Example Task

```
{  
  "resourceType": "Task",  
  "id": "task-medication-1",  
  "status": "ready",  
  "intent": "order",  
  "code": {  
    "text": "Administer Antibiotic"  
  },  
  "for": {  
    "reference": "Patient/patient-1"  
  }  
}
```

```
    },  
    "encounter": {  
      "reference": "Encounter/encounter-admission-1"  
    }  
  }  
}
```

---

## STEP 12 — Medication Management

### Resources

- Medication
- MedicationRequest

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## MedicationRequest Example

```
{  
  "resourceType": "MedicationRequest",  
  "id": "medreq-1",  
  "status": "active",  
  "intent": "order",  
  "medicationCodeableConcept": {  
    "text": "Ceftriaxone 1g"  
  },  
  "subject": {  
    "reference": "Patient/patient-1"  
  },  
  "encounter": {  
    "reference": "Encounter/encounter-admission-1"  
  }  
}
```

---

## IMPORTANT RECOMMENDATION

For production:

Implement:

```
MedicationAdministration
```

Because:

```
MedicationRequest = Order  
MedicationAdministration = Actual administration
```

---

## STEP 13 — Monitoring & Vitals

Inpatients require continuous monitoring.

### Resource

- Observation

Examples:

- BP
- Pulse
- SPO2
- Temperature
- Intake/output
- Pain score

---

## Example

```
{  
  "resourceType": "Observation",  
  "id": "obs-spo2-1",  
  "status": "final",  
  "code": {  
    "text": "Oxygen Saturation"  
  },  
  "subject": {  
    "reference": "Patient/patient-1"  
  },  
  "valueQuantity": {  
    "value": 95,  
    "unit": "%"  
  }  
}
```

```
}  
}
```

## STEP 14 — Diagnostic Workflow

Resources:

- Observation
- DiagnosticReport
- ServiceRequest

## Example Flow

```
Doctor Orders MRI  
  ↓  
ServiceRequest  
  ↓  
Radiology Task  
  ↓  
DiagnosticReport Generated  
  ↓  
Doctor Review Task
```

## STEP 15 — Procedure Workflow

Examples:

- Surgery
- Intubation
- Central line insertion
- Dialysis

## Procedure Example

```
{  
  "resourceType": "Procedure",
```

```
"id": "procedure-1",
"status": "completed",
"code": {
  "text": "Appendectomy"
},
"subject": {
  "reference": "Patient/patient-1"
},
"encounter": {
  "reference": "Encounter/encounter-admission-1"
}
}
```

---

## STEP 16 — ICU / Ward Transfer

VERY IMPORTANT inpatient workflow.

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### Example

```
Ward
  ↓
ICU
  ↓
Ward
```

---

## Use Encounter + Location

### Example

```
{
  "location": [
    {
      "location": {
        "reference": "Location/icu-bed-1"
      },
      "status": "active"
    }
  ]
}
```

```
]
}
```

---

## Recommended Transfer Task

```
{
  "resourceType": "Task",
  "id": "task-transfer-1",
  "status": "requested",
  "intent": "order",
  "code": {
    "text": "Transfer Patient to ICU"
  }
}
```

---

## STEP 17 — Clinical Documentation

Strong recommendation:

Use:

```
DocumentReference
```

Examples:

- progress notes
- surgery notes
- discharge summaries
- scanned reports
- ICU notes
- consent forms

---

## Example

```
{
  "resourceType": "DocumentReference",
  "id": "doc-progress-note-1",
```

```
"status": "current",
"type": {
  "text": "Progress Note"
},
"subject": {
  "reference": "Patient/patient-1"
},
"content": [
  {
    "attachment": {
      "contentType": "application/pdf",
      "url": "https://files.example.com/progress-note.pdf"
    }
  }
]
}
```

---

## STEP 18 — AI Workflow Opportunities

Inpatient systems are VERY suitable for AI.

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### AI Use Cases

| AI Agent             | Purpose                 |
|----------------------|-------------------------|
| Monitoring Agent     | Detect deterioration    |
| Documentation Agent  | Generate summaries      |
| Coding Agent         | ICD/CPT suggestions     |
| Risk Agent           | Sepsis/fall risk        |
| Bed Management Agent | Optimize bed allocation |
| Discharge Agent      | Predict discharge       |
| Pharmacy Agent       | Drug interaction checks |

---

## Example AI Workflow

```
Vitals Updated
  ↓
AI Monitoring Agent
  ↓
Detects abnormal SP02 trend
  ↓
Creates Urgent Task
  ↓
Doctor Alert
```

---

## STEP 19 — Discharge Planning

Discharge planning often starts EARLY.

Resources:

- CarePlan
- Task
- DocumentReference

---

## Example Tasks

```
Physiotherapy Evaluation
Medication Counseling
Insurance Clearance
Discharge Summary Preparation
```

---

## STEP 20 — Generate Discharge Summary

Strong recommendation:

Store as:

DocumentReference

## Example

```
{
  "resourceType": "DocumentReference",
  "id": "doc-discharge-1",
  "status": "current",
  "type": {
    "text": "Discharge Summary"
  },
  "subject": {
    "reference": "Patient/patient-1"
  },
  "context": {
    "encounter": [
      {
        "reference": "Encounter/encounter-admission-1"
      }
    ]
  }
}
```

## STEP 21 — Billing Workflow

Resources:

- Invoice
- Claim
- ClaimResponse

## IMPORTANT

Inpatient billing is MUCH more complex.

Includes:

- bed charges

- nursing charges
  - ICU charges
  - procedures
  - labs
  - imaging
  - medications
  - consumables
- 

## Invoice Example

```
{
  "resourceType": "Invoice",
  "id": "invoice-1",
  "status": "issued",
  "subject": {
    "reference": "Patient/patient-1"
  },
  "totalNet": {
    "value": 185000,
    "currency": "INR"
  }
}
```

---

## STEP 22 — Insurance Workflow

### Claim Example

```
{
  "resourceType": "Claim",
  "id": "claim-1",
  "status": "active",
  "patient": {
    "reference": "Patient/patient-1"
  }
}
```

---

## STEP 23 — Discharge

Encounter becomes:

```
"status": "finished"
```

EpisodeOfCare becomes:

```
"status": "finished"
```

---

## COMPLETE INPATIENT FLOW

```
Patient Registration
↓
Coverage Verification
↓
Admission Decision
↓
EpisodeOfCare Created
↓
Bed Allocation
↓
Admission Encounter
↓
Nursing Assessment
↓
Doctor Evaluation
↓
Orders
↓
Medication Administration
↓
Monitoring
↓
Transfers
↓
Procedures
↓
Progress Notes
↓
Discharge Planning
```

↓  
Invoice  
↓  
Claim  
↓  
Discharge Summary  
↓  
Discharge

---

## IMPORTANT TASK WORKFLOWS

### Recommended Task Categories

| Category   | Purpose                   |
|------------|---------------------------|
| admission  | Admission processing      |
| nursing    | Nursing workflows         |
| medication | Medication administration |
| lab        | Lab workflows             |
| radiology  | Imaging workflows         |
| transfer   | Bed/ward transfer         |
| surgery    | OR workflows              |
| discharge  | Discharge planning        |
| billing    | Billing workflows         |
| insurance  | Claims processing         |

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## BED MANAGEMENT ARCHITECTURE

VERY IMPORTANT.

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## Recommended Location Types

Hospital  
Building  
Floor  
Ward  
Room  
Bed  
ICU Bed  
Operation Theater

---

## Recommended Bed States

available  
occupied  
cleaning  
reserved  
maintenance  
blocked

---

## RECOMMENDED AI WORKFLOWS

### 1. Deterioration Detection

Vitals trend analysis

---

### 2. Bed Optimization

Predict discharges  
Suggest bed assignments

---

### 3. Clinical Summaries

Generate progress notes  
Generate discharge summaries

### 4. Insurance Automation

Predict denials  
Generate claims  
Validate documentation

## RECOMMENDED EVENT-DRIVEN ARCHITECTURE

### Suggested Events

patient.admitted  
bed.assigned  
encounter.started  
medication.administered  
vitals.recorded  
procedure.completed  
patient.transferred  
patient.discharged  
invoice.generated  
claim.submitted

## RECOMMENDED PRODUCTION RESOURCE SET

Strongly recommended inpatient resources:

Patient  
Coverage  
Encounter  
EpisodeOfCare

Location  
Task  
Observation  
Condition  
CarePlan  
ServiceRequest  
Procedure  
Medication  
MedicationRequest  
MedicationAdministration  
DiagnosticReport  
DocumentReference  
Invoice  
Claim  
ClaimResponse  
AllergyIntolerance  
Provenance  
AuditEvent

---

## IMPORTANT ARCHITECTURAL PRINCIPLES

### 1. EpisodeOfCare = Hospitalization

One hospitalization  
=  
One EpisodeOfCare

---

### 2. Encounter = Clinical Interaction

Admission  
Ward  
ICU  
Transfer  
Discharge

### 3. Location = Physical Tracking

Tracks:

- bed
  - ward
  - ICU
  - OT
  - room
- 

### 4. Task = Workflow Orchestration

Tracks:

- nursing
  - medications
  - transfers
  - billing
  - discharge
- 

### 5. DocumentReference = Clinical Documents

Stores:

- discharge summaries
  - progress notes
  - scanned reports
  - surgical notes
- 

## FINAL ARCHITECTURE RECOMMENDATION

```
FHIR Resources = Clinical Truth
EpisodeOfCare = Hospitalization Context
Task = Workflow Engine
Location = Physical Hospital Tracking
AI Agents = Automation Layer
Events = Communication Layer
```

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# FINAL SUMMARY

Inpatient workflow is MUCH more complex than outpatient.

The most important additions are:

1. EpisodeOfCare
2. Location
3. Task
4. CarePlan
5. MedicationAdministration
6. Coverage
7. DocumentReference
8. AI Monitoring Workflows

The best architecture is:

FHIR + Task + Events + AI + Location + EpisodeOfCare

This architecture scales well for:

- enterprise hospitals
- ICU systems
- nursing workflows
- surgery workflows
- AI monitoring systems
- insurance workflows
- bed management systems
- operational healthcare platforms